Curriculum Vitae



Nagendra S Kamath

Indian Institute of Technology Mandi, India
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EDUCATION

Ph.D. in Physics *Feb 2022 – Present*

School of Physical Sciences, Indian Institute of Technology Mandi CGPA: **8.25/10**

Thesis Title: Effects of Substitutional Doping on Ultrafast Many-Body Interactions and Valley Dynamics in Two-Dimensional Transition Metal Dichalcogenides

Thesis Advisor: Prof. Suman Kalyan Pal

M.Sc. in Physics *Aug 2018 – Oct 2020*

Department of Studies in Physics, University of Mysore CGPA: **8.38/10**

B.Sc. in Physics, Chemistry and Mathematics *Aug 2015 – Jun 2018*

Yuvaraja's College, University of Mysore Percentage: **82.08%**

PROFESSIONAL EXPERIENCE

Project Associate - I *Apr 2021 - Feb 2022*

Indian Institute of Technology Mandi, Himachal Pradesh

- **Project Title**: Study the dynamical evolution of spin and valley related many particle electronic states in two dimensional transition metal dichalcogenides using ultrafast time-resolved spectroscopy.

RESEARCH SUMMARY

Ph.D. Scholar, Indian Institute of Technology Mandi *Feb 2022 – Present*

- Developed microscopic TA spectroscopy capabilities for site-selective ultrafast optical measurements in 2D materials.
- Developed a polarization-resolved micro-photoluminescence (micro-PL) setup for investigating valley polarization in 2D materials.
- Helped establish a low-temperature TA, helicity-resolved TA (HRTA) measurement capabilities for investigating temperature-dependent carrier dynamics and polarization-dependent measurements.
- Contributed to the development of the laboratory's terahertz (THz) spectroscopy system from the gound up.
- Selected under the NFFA-Europe programme to conduct helicity-resolved photoluminescence (HRPL) measurements at IESL-FORTH, Greece, to investigate valley polarization in 2D materials.

Project Associate - I, Indian Institute of Technology Mandi *Apr 2021 - Feb 2022*

- Optimized mechanical exfoliation protocols for monolayer 2D TMD samples.

M.Sc. Student, University of Mysore *Aug 2018 - Oct 2020*

- Contributed to two studies on structure–property correlations in millets and areca nuts using XRD, Raman, SEM, FTIR and AC susceptibility techniques.

RESEARCH PUBLICATIONS

[1] **Formation of charged defect excitons and reconfigured exciton relaxation pathways in chromium-doped monolayer MoS_2, N. S. Kamath**, L. Saran, D. Maity, T. N. Narayanan, S. K. Pal, ***Nanoscale***, *2026*

[2] **Ultrafast Photoinduced Charge Separation in Quasi-2D Perovskites Enables Enhanced Photocurrent Generation**, S. Halder, K. Gayen, **N. S. Kamath**, U. Singh, S. Ghosal, S. K. Pati, S. K. Pal, ***Advanced Optical Materials***, *e71508*, *2026*

[3] **Enhancing the Performance of Perovskite Solar Cells through Interfacial Modification by Fluorinated 2D Perovskite**, K. Gayen, P. Kumar, **N. S. Kamath**, R. Singh, S. K. Pal, ***ACS Applied Energy Materials***, *9 (11), 6793-6803*, *2026*

[4] **Spin-Selective Defect Coupling Drives Valley Polarization and Dynamics in Vanadium-Doped Monolayer MoS_2**, **N. S. Kamath**, A. Soni, D. Maity, T. N. Narayanan, S. K. Pal, ***The Journal of Physical Chemistry C***, *130 (12), 4504-4513*, *2026*

[5] **Valley Properties and Dynamics in 2D Transition Metal Dichalcogenides: Prospects for Valleytronic Applications**, **N. S. Kamath**, A. Soni, S. K. Pal, ***The Journal of Physical Chemistry Letters*** *(Accepted)*, *2025*

[6] **Microwave-induced plasma drilling of kenaf-reinforced polypropylene composites: Effects of tool material on reusability and hole quality via optical emission spectroscopy**, Rampal, **N. S. Kamath**, S. Zafar, ***Journal of Manufacturing Processes***, *155, 589-605*, *2025*

[7] **Valley Engineering of Monolayer MoS_2 via Substitutional Cr and V Dopants and the Mechanistic Insights**, S. Anil, D. Maity, S. Adabala, A. De, **N. S. Kamath**, R. Hammad, J. J. Panda, R. K. Biroju, V. Vretenar, Rajalakshmi G, S. K. Pal, S. Ghosh, T. N. Narayanan, ***Nano Letters***, *25, 13401-13409*, *2025*

[8] **Enhanced valley polarization in MoS_2 via substrate-induced strain and transition dipole moment modulation**, A. Soni, **N. S. Kamath**, Y. Shen, W. Chang, A. D. Sarkar, S. K. Pal, ***Nanoscale***, *17, 25727-25739*, *2025*

[9] **Substrate-induced modulation of transient optical response of large-area monolayer MoS_2**, A. Soni, **N. S. Kamath**, Y. Shen, H. Seksaria, A. D. Sarkar, W. Chang, S. K. Pal, ***Scientific Reports***, *15, 7537*, *2025*

[10] **Exciton–phonon coupling in quasi-two-dimensional Ruddlesden–Popper perovskites: impact of a mixed-phase structure**, S. Sarkar, **N. S. Kamath**, K. Gayen, S. K. Pal, ***Nanoscale***, *17, 10771-10783*, *2025*

[11] **Charge Transfer Mediated Photoluminescence Engineering in WS_2 Monolayers for Optoelectronic Application**, B. Luhar, R. Dhankhar, R. V. Nair, A. Soni, **N. S. Kamath**, S. K. Pal, V. Balakrishnan, ***ACS Applied Nano Materials***, *7 (19), 22350-22359*, *2024*

Book Chapter

[1] **Photophysics and Excited-State Dynamics of 2D Quantum Materials for Energy Conversion**, N. S. Kamath, A. Soni, A. S. Sarkar, S. K. Pal, ***Handbook of Two-Dimensional Quantum Materials and Energy Applications, CRC Press***, *2026*

CONFERENCES AND SEMINARS

- Poster Presentation - Graphene India 2026 - Kochi - March 09-12, 2026. **(Best Poster Award)**
- Poster Presentation - 12th Theme meeting on Ultrafast Sciences (UFS) 2025 - IIT Hyderabad - November 13-15, 2025.
- Poster Presentation - International conference - 2D Transition Metal Dichalcogenides 2025 at Churchill College, University of Cambridge, July 20-24, 2025.
- Poster Presentation - International conference on Ultrafast Nonlinear Optics and Optical Spectroscopy (UNOOS) - IISER Mohali - 10-12 Dec 2024. **(Best Poster Award)**
- Poster Presentation - One Day Workshop - Electronic Materials: Preparation, Characterization and Application - IIT Mandi - 20 Apr 2024.
- Poster Presentation - One Day Workshop: Celebrating National Science Day, School of Physical Sciences, Indian Institute of Technology Mandi, 28 Feb 2024.

- Poster Presentation - 10th Theme meeting on Ultrafast Sciences (UFS) 2023 - NPL Delhi - 25-27 Nov 2023.
- Attended Virtual Online Workshop - Light-Matter Interactions in Low Dimensional and Topological Photonic Materials - IISc Bangalore - 27 Jan - 1 Feb 2022.

HONORS AND AWARDS

ANRF International Travel Support (ITS) Grant *July 2025*

- Awarded by the Anusandhan National Research Foundation (ANRF), Government of India, to support international travel for presenting research work at a scientific conference.

INSPIRE Fellowship *2022 - 2027*

- Awarded by the Department of Science and Technology, Govt. of India, recognizing the academic excellence and research potential in the field of Physics.

KSTA Merit Cash Award *2015*

- Awarded by the Karnataka Science and Technology Academy for academic excellence.

INSPIRE-SHE Scholarship *2015 - 2020*

- Awarded by the Department of Science and Technology, Govt. of India, for ranking among the top 1% in class XII.

Best Outgoing Student of the Year *2013*

- Awarded by Janata Vidyalaya Dandeli in recognition of outstanding academic performance and overall excellence.

RESEARCH SKILLS

- **Ultrafast Spectroscopy:** Extensive experience in transient absorption (TA) spectroscopy, including microscopic TA, low-temperature TA, and helicity-resolved TA (HRTA); experience in ultrafast laser operation, maintenance and upkeep.
- **Photoluminescence Spectroscopy:** Temperature-dependent and polarization-resolved photoluminescence (PL) spectroscopy; routine calibration and maintenance of PL instrumentation.
- **Raman Spectroscopy:** Operation of the Horiba LabRAM HR Evolution Raman spectrometer; experience as a Teaching Assistant for the instrument at the Advanced Materials Research Centre, IIT Mandi.
- **Material Preparation:** Mechanical exfoliation of layered two-dimensional materials and preparation of monolayer TMD samples.
- **Additional Instrumentation:** Instrumentation-level setup and operation of Z-scan and terahertz (THz) spectroscopy systems.

ADDITIONAL SKILLS AND INTERESTS

Technical Skills C | Python | Matlab | GNU Octave | LaTeX

Languages ***Fluent:*** Kannada English Hindi Konkani ***Beginner:*** Japanese French

Hobbies Deltiology | Chess | Trivia

REFERENCES

- **Prof. Suman Kalyan Pal** (Ph.D. Supervisor)

Professor, School of Physical Sciences, Indian Institute of Technology Mandi

Email: suman@iitmandi.ac.in

- **Prof. Kaustav Mukherjee** (Doctoral committee head)

Professor, School of Physical Sciences, Indian Institute of Technology Mandi

Email: kaustav@iitmandi.ac.in

Declaration

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.